Harder Geometrical Problems — problem 12.3.2.47

A. Engel — Problem-Solving Strategies · 1998

A circle with center O, which is inscribed into $\triangle ABC$, touches its sides in A_1, B_1, C_1. The segments AO, BO, CO intersect the circle in A_2, B_2, C_2, respectively. Prove that A_1A_2, B_1B_2, C_1C_2 intersect in one point.